

Stub Match & Smith Chart Questions

9. Plot the point on the Smith chart given by a short circuit stub of zero length.
10. Now plot this point as an admittance.
11. Plot the original load $Z_L = 35 + 55j \Omega$ as an admittance.
12. Repeat Q7-8 but using a parallel rather than a series stub match.
13. Why is a double stub match often a more practical solution than a single stub match?
14. You find your board for your series stub match has had the stub added at the wrong point 0.25λ from the load. You have some copper tape. How might you still achieve a match?
15. For the load $Z_L = 10 + 20j \Omega$ you are given a connecting line ($Z_0 = 50$) with a double stub match 0.04λ from the load. The stubs are spaced $3/8\lambda$ apart. What type & length do the stubs need to be to provide a match?